

AUTOMOTIVE OPEN SYSTEM ARCHITECTURE (AUTOSAR)

High-Level Design Specification Body Control Module (BCM)

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NOTICE: This High-Level Design document contains synthetic architectural models designed for automated retrieval, traceability validation, and inconsistency reporting. All components, ports, and signals comply with Classic AUTOSAR 4.4 methodology. Advisory output requires engineer review.

Document Control & Revision History

Section 0.1: Change History Log

Rev	Date	Author	Change Description	Approval Status
0.1	2026-01-10	Arch Team	Initial draft structure and component boundaries	Draft
0.5	2026-02-04	Systems Lead	Added Port/Interface tables and signal cycle times	Reviewed
1.0	2026-03-15	Chief Architect	Formal baseline release v1.0. All 10 SWCs frozen.	Approved
2.0	2026-08-20	Systems Lead	ECN-2026-08: Signal renaming, interface refactoring, and defect injection test revision.	Pending Review

Section 0.2: Approvals & Signatures

Role	Name	Organization	Sign Date
Lead Systems Architect	Dr. S. K. Kulkarni	PCCOE Automotive Lab	2026-03-16
Safety Governance Lead	Prof. V. Sharma	Functional Safety Board	2026-03-17
Diagnostic Verification Lead	Eng. R. Nair	Validation Group	2026-03-18

1. Introduction and Architectural Scope

1.1 Purpose of the High-Level Design

This High-Level Design (HLD) document defines the software architecture, component boundaries, port interfaces, signals, and interaction flows for the Body Control Module (BCM ECU). The BCM is responsible for managing core cabin functions including central door locking, power window automation, exterior lighting schemes, windshield wiper/washer cycles, immobilizer authentication, and diagnostic event recording. The architecture is formalized under Classic AUTOSAR Release 4.4.0 and targets deployment on a 32-bit automotive microcontroller with dual CAN-FD transceivers and a LIN sub-bus master controller.

1.2 System Scope & Operational Boundaries

The scope of this document encompasses all Application Software Components (Application SWCs) located above the AUTOSAR Runtime Environment (RTE). Basic Software (BSW) services such as the Diagnostic Event Manager (Dem), Diagnostic Communication Manager (Dcm), CAN State Manager (CanSM), and Operating System (OS) are referenced via standardized Service Ports and BSW bridges.

In-Scope Functional Domains:

- Central Door Locking, child lock supervision, and crash-triggered emergency unlocking.
- Smart Power Window control with express motion and anti-pinch safety algorithms.
- Exterior Lighting (headlamps, turn signals, hazard lights) and ambient illumination.
- Rain-sensing wiper intervals, high/low speed control, and washer fluid pump cycling.
- Power state transitions between Active, Standby, and Low Voltage Sleep modes.

Out-of-Scope Domains:

- Powertrain management, engine start sequence, and high-voltage battery charging.
- Hardware PCB layout schematics and low-level MCAL register definitions.

1.3 Architectural Standards & Compliance

All software specifications in this document adhere to the following normative automotive standards:

1. **AUTOSAR CP R4.4.0:** Component model, virtual function bus concepts, and standardized interface templates.
2. **ISO 26262:2018:** Road Vehicles - Functional Safety (governing safety integrity levels up to ASIL-D).
3. **ISO 14229-1 (UDS):** Unified Diagnostic Services on Controller Area Network.
4. **ISO 11898-1/2:** Controller Area Network (CAN-FD) data link layer and physical transmission.

2. System Context and Hardware Boundary

2.1 BCM Physical Context

The BCM ECU acts as the primary gateway and actuator control center for vehicular body electronics. It interfaces directly with the central vehicle gateway over a 500 kbps / 2 Mbps CAN-FD network (Channel 1) and commands smart sub-actuators and sensors via a 19.2 kbps LIN sub-bus (Channel 0).

Physical Transceivers & Interfaces:

- Dual high-speed CAN-FD transceivers supporting ISO 11898-2 with wake-up over bus capability.
- Single-wire LIN master transceiver compliant with LIN 2.2A.
- High-Side Smart Drivers (PROFET) for direct actuation of exterior lamps, wiper motors, and latch coils.
- Configurable analog/digital input channels with low-pass hardware filtering for door switches.

2.2 ECU Power Supply & Operating States

The module operates under nominal 12V automotive battery voltage (operating range: 9.0V to 16.0V DC). Extended crank operating capability is maintained down to 6.0V DC for 500ms. Overvoltage protection isolates actuator stages when battery voltage exceeds 18.0V DC.

Power Modes:

1. **RUN Mode (Active):** Microcontroller at 80 MHz, all SWC runnables executed periodically, full CAN/LIN transmission.
2. **STANDBY Mode:** Core clock reduced to 8 MHz, actuators unpowered, CAN transceiver listening for network management frames.
3. **SLEEP Mode:** Deep quiescent state (< 100 μ A current draw), wake-up initiated only via keyless remote or door handle pull switch.

2.3 Safety Concept & ASIL Allocation Matrix

Function / SWC	Hazard Scenario	Safety Goal	ASIL
DoorLockSWC	Doors remain locked during collision	Unlock all latches within 30ms of crash detection	ASIL-B
WindowCtrlSWC	Pinch injury during automatic window close	Reverse motor within 50ms upon obstacle force > 100N	ASIL-A
LightingMgrSWC	Total loss of forward illumination at night	Maintain low beam or activate backup parking lamp	ASIL-B
WiperWasherSWC	Wiper stalls during heavy rain at speed	Execute limp-home slow sweep mode	ASIL-A
ImmobilizerSWC	Unauthorized engine start without valid key	Prevent immobilizer token release	QM

3. AUTOSAR Architecture Overview & 4. Software Components

3.1 Layered Structure & Virtual Function Bus (VFB)

The BCM architecture is partitioned into four primary AUTOSAR layers: Application Layer (10 Application SWCs), Runtime Environment (RTE), Basic Software (BSW), and Microcontroller Abstraction Layer (MCAL). All inter-SWC communication flows strictly through formal Ports and Interfaces across the RTE.

4.1 Software Component (SWC) Inventory

The system defines 10 core Application Software Components:

Component Name	Type	ASIL	Period	Functional Description
DoorLockSWC	Application	ASIL-B	20 ms	Central door locking, child safety, auto-lock at 15 km/h, and emergency crash unlock override.
WindowCtrlSWC	Application	ASIL-A	20 ms	Power window motor positioning, express movement, and anti-pinch safety reverse algorithms.
LightingMgrSWC	Application	ASIL-B	20 ms	Headlamps, turn signals, hazard emergency flashers, and interior ambient lighting.
WiperWasherSWC	Application	ASIL-A	50 ms	Rain sensor processing, intermittent wiper pacing, and washer pump motor control.
ImmobilizerSWC	Application	QM	100 ms	Transponder key challenge-response cryptographic authentication with engine ECM.
SeatAdjustmentSWC	Application	QM	100 ms	Electric seat adjustment motors, Hall sensor feedback, and user memory presets.
CanComSWC	Service Bridge	ASIL-B	10 ms	CAN-FD frame serialization, signal filtering, E2E protection, and network management.
LinComSWC	Service Bridge	QM	20 ms	LIN master coordinator for rain sensor, mirror sub-nodes, and smart ambient LEDs.
DiagMgrSWC	Diagnostics	ASIL-B	100 ms	UDS ISO 14229 services, DTC fault logging, and freeze-frame snapshot capture.
PowerMgrSWC	Mode Control	ASIL-B	20 ms	ECU operational modes (Active, Standby, Sleep) and battery low-voltage load shedding.

4.2 Auxiliary System Dependencies

Note on Extended Cabin Modules: In this revision, the rain-closure sequence delegates roof actuation to **SunroofCtrlSWC** (Refer to Section 7 for dependency specifications).

5. Port and Interface Catalog

5.1 Sender-Receiver (S/R) Interfaces

The table below specifies all Sender-Receiver interfaces, data elements, data types, and units:

Interface Name	Data Element	Data Type	Range / Enum	Unit
If_DoorLockStatus	DoorLockState	UInt8	0:UNLOCKED, 1:LOCKED, 2:DEADLOCK	Enum
If_DoorLockStatus	ChildLockActive	Boolean	0:False, 1:True	Flag
If_WindowState	WindowPosition_FL	UInt8	0 to 100	% Open
If_WindowState	AntiPinchTriggered	Boolean	0:False, 1:True	Flag
If_LightingCommand	ExteriorLightMode	UInt8	0:OFF, 1:PARK, 2:LOW, 3:HIGH	Enum
If_LightingCommand	HazardActive	Boolean	0:False, 1:True	Flag
If_VehicleSpeed	VehicleSpeedKmh	Float32	0 to 350	km/h
If_BatteryVoltage	SupplyVoltage_mV	UInt16	0 to 32000	mV
If_BatteryVoltage	LowVoltageWarning	Boolean	0:False, 1:True	Flag
If_WiperSpeedCmd	WiperSpeedMode	UInt8	0:OFF, 1:INT1, 2:INT2, 3:LOW, 4:HIGH	Enum
If_RainIntensity	RainSensorLevel	UInt8	0 to 255	Raw Lux
If_CrashStatus	CrashDetected	Boolean	0:No Crash, 1:Crash Event	Flag
If_TrailerHitchDetect	TrailerAttached	Boolean	0:No Trailer, 1:Attached	Flag

5.2 Client-Server (C/S) Interfaces

Interface Name	Operation	Argument (Direction: Type)	Return Type
If_AuthSecurity	AuthenticateKey	IN seed: UInt32, OUT key: UInt32	Std_ReturnType
If_DiagDtcService	SetDTC	IN dtcId: UInt32, IN status: UInt8	Std_ReturnType
If_PowerModeControl	RequestPowerState	IN targetState: UInt8, OUT ack: Boolean	Std_ReturnType

5.3 Core Component Functional Profiles

Profiles for DoorLockSWC, WindowCtrlSWC, LightingMgrSWC, and DiagMgrSWC:

DoorLockSWC (Safety: ASIL-B)

Functional Behavior: Central door locking, auto-lock above 15 km/h, and emergency crash unlock within 30ms.

Runnable Entities: Runnable_DoorLock_20ms (Cyclic evaluation), Runnable_DoorLock_CrashEvent (Event interrupt).

WindowCtrlSWC (Safety: ASIL-A)

Functional Behavior: Express power window motion and anti-pinch safety reversal upon force > 100N.

Runnable Entities: Runnable_WindowCtrl_20ms (Positioning loop), Runnable_PinchDetect_10ms (Current feedback).

LightingMgrSWC (Safety: ASIL-B)

Functional Behavior: Exterior headlamps, turn signal flashers, and emergency hazard flashers at 1.5 Hz.

Runnable Entities: Runnable_Lighting_20ms (Arbitration), Runnable_HazardFlash_250ms (Cadence square-wave).

DiagMgrSWC (Safety: ASIL-B)

Functional Behavior: UDS ISO 14229 services, DTC fault logging, and non-volatile freeze-frame records.

Runnable Entities: Runnable_Diag_100ms (Debounce counters), Runnable_UdsService_Handler (Diagnostic messages).

6. Signal List, Data Mapping and Bus Allocation

6.1 CAN-FD & LIN Signal Table

The following table details every network signal mapped to CAN-FD (Channel 1) or LIN (Channel 0):

Signal Name	Type	Cycle	Bus	Source SWC	Destination SWC
Sig_DoorState_FL_Status	Uint8	20 ms	CAN-FD	CanComSWC	DoorLockSWC
Sig_AutoLockSpeedThreshold	Uint16	100 ms	CAN-FD	CanComSWC	DoorLockSWC
Sig_CrashNotification	Boolean	10 ms	CAN-FD	CanComSWC	DoorLockSWC
Sig_ExteriorHeadlamp_State	Uint8	50 ms	CAN-FD	LightingMgrSWC	CanComSWC
Sig_RainSensorVal	Uint8	100 ms	LIN	LinComSWC	WiperWasherSWC
Sig_WindowObstacleDetected	Boolean	20 ms	Internal	WindowCtrlSWC	DiagMgrSWC
Sig_BatterySupplyMV	Uint16	1000 ms	Internal	PowerMgrSWC	DiagMgrSWC
Sig_IgnitionSwitchPos	Uint8	20 ms	CAN-FD	CanComSWC	PowerMgrSWC
Sig_TurnSignalRequest	Uint8	50 ms	CAN-FD	LightingMgrSWC	CanComSWC
Sig_SeatMemoryPresetIndex	Uint8	200 ms	CAN-FD	SeatAdjustmentSWC	CanComSWC
Sig_WiperParkPosition	Boolean	50 ms	Internal	WiperWasherSWC	CanComSWC
Sig_AmbientBrightnessLux	Uint16	500 ms	LIN	LinComSWC	LightingMgrSWC

7. Component Dependencies and Port Connections

7.1 Component Port Bindings

The following table details Provided Ports (PPorts) and Required Ports (RPorts) across key SWCs:

Software Component	Provided Ports (PPort)	Required Ports (RPort)
DoorLockSWC	[REMOVED IN v2]	r_VehicleSpeed (If_VehicleSpeed), r_CrashStatus (If_CrashStatus), r_PowerState (If_PowerModeControl)
WindowCtrlSWC	p_WindowState (If_WindowState)	r_VehicleSpeed (If_VehicleSpeed), r_PowerState (If_PowerModeControl)
LightingMgrSWC	p_LightingCommand (If_LightingCommand), p_TrailerHitch (If_TrailerHitchDetect)	r_PowerState (If_PowerModeControl), r_CrashStatus (If_CrashStatus)
WiperWasherSWC	p_WiperSpeedCmd (If_WiperSpeedCmd)	r_VehicleSpeed (If_VehicleSpeed), r_RainIntensity (If_RainIntensity)
CanComSWC	p_VehicleSpeed (If_VehicleSpeed), p_CrashStatus (If_CrashStatus)	r_DoorLockStatus (If_DoorLockStatus), r_LightingCommand (If_LightingCommand)
PowerMgrSWC	p_PowerState (If_PowerModeControl), p_BatteryVoltage (If_BatteryVoltage)	r_IgnitionSwitch (If_VehicleSpeed)
DiagMgrSWC	p_DiagDtcService (If_DiagDtcService)	r_BatteryVoltage (If_BatteryVoltage)
ImmobilizerSWC	p_AuthSecurity (If_AuthSecurity)	r_PowerState (If_PowerModeControl)

7.2 Coupling Constraints & Power Shedding Rules

- Safety Priority:** DoorLockSWC and LightingMgrSWC must prioritize signals originating from p_CrashStatus above all convenience and user switch requests.
- Coupling Rules:** Inter-SWC communication flows exclusively through intermediate RTE buffers.
- Power Shedding:** If SupplyVoltage_mV drops below 10,500 mV, PowerMgrSWC sheds seat heaters, power windows, and ambient lighting sequentially.

8. Hardware Interface & Connector Pinout

8.1 Physical Connector Pin Allocations

The table below details physical connector allocations for the BCM ECU:

Pin ID	Connector	Signal Name	Electrical Spec	Fail-Safe Default	ASIL
PIN_01	CN-A	Sig_DoorState_FL_Status	0-12V Digital Input	HIGH_IMPEDANCE	ASIL-B
PIN_02	CN-A	Sig_DoorState_FrontRight	0-12V Digital Input	HIGH_IMPEDANCE	ASIL-B
PIN_03	CN-A	Sig_EmergencyCrashPulse	0-5V TTL Interrupt	PULL_HIGH (TRIP)	ASIL-D
PIN_04	CN-B	Sig_CanHigh_HS_Ch1	CAN-FD ISO 11898-2	RECESSIVE	ASIL-C
PIN_05	CN-B	Sig_CanLow_HS_Ch1	CAN-FD ISO 11898-2	RECESSIVE	ASIL-C
PIN_06	CN-C	Sig_LinBusMaster_Ch0	LIN 2.2A Single Wire	RECESSIVE	QM
PIN_07	CN-C	Sig_RainSensorVal_Pin	PWM 100Hz 5V	DUTY_50_DEFAULT	QM
PIN_08	CN-D	Sig_HeadlampRelayHigh	12V 25A Driver	FORCE_OFF	ASIL-B
PIN_09	CN-D	Sig_HazardFlasherDirect	12V 15A Pulsed	FORCE_FLASHING	ASIL-C
PIN_10	CN-E	Sig_WiperParkSwitch	Active-Low Ground	WIPER_HOME	ASIL-A
PIN_11	CN-E	Sig_WindowPinchSense	Current 0-20mA	STOP_REVERSE	ASIL-B
PIN_12	CN-F	Sig_VBat_SupplyMonitor	Analog ADC 0-32V	NOMINAL_12V	ASIL-B

9. Functional Flow Sequences

9.1 Flow 1: Automatic Speed-Dependent Door Locking (Auto-Lock)

Precondition: Vehicle engine is running, passenger doors closed, initial speed 0 km/h.

Execution Steps:

1. Wheel speed sensors transmit periodic pulses to ABS/ESP ECU.
2. ESP ECU broadcasts velocity on CAN-FD Channel 1.
3. CanComSWC unpacks CAN frame and writes VehicleSpeedKmh to RTE port p_VehicleSpeed.
4. DoorLockSWC evaluates r_VehicleSpeed at 20ms invocation period.
5. When VehicleSpeedKmh exceeds 15 km/h:
 - a. DoorLockSWC asserts output latch driver signals for all 4 doors simultaneously.
 - b. Hardware solenoids energize for 120ms driving latches to LOCKED position.
 - c. DoorLockSWC updates DoorLockState = 1 (LOCKED) on port p_DoorLockStatus.
6. Cluster display receives lock status confirmation and activates icon.

9.2 Flow 2: Crash Hazard Unlock and Lighting Response

Precondition: Vehicle is in motion or stationary; door latches are in LOCKED position.

Execution Steps:

1. Airbag Restraint Control Module (RCM) detects crash deceleration exceeding 5g.
2. RCM asserts discrete hardwired interrupt line PIN_03 and broadcasts Crash CAN-FD frame.
3. CanComSWC asserts p_CrashStatus.CrashDetected = TRUE within < 5ms.
4. DoorLockSWC intercepts crash event via r_CrashStatus:
 - a. Overrides manual lock switch commands and child lock inhibitions.
 - b. Pulses unlock solenoid drivers for 250ms to force latches into UNLOCKED state.
5. LightingMgrSWC receives emergency trigger:
 - a. Immediately engages hazard flasher relays at continuous 1.5 Hz cadence.
 - b. Activates interior ambient courtesy lamps at 100% brightness.
6. DiagMgrSWC registers crash freeze-frame record and locks DTC B10A2 into EEPROM.

10. Diagnostic Specifications & DTC Matrix

10.1 UDS ISO 14229 Service Support

The BCM diagnostic stack implements Unified Diagnostic Services (UDS) over ISO-TP:

- **Service 0x10:** Diagnostic Session Control (Default, Programming, Extended Sessions).
- **Service 0x14:** Clear Diagnostic Information (Group 0xFFFFFFFF).
- **Service 0x19:** Read DTC Information (sub-functions 0x02 reportByStatusMask, 0x06 reportSnapshotRecord).
- **Service 0x22:** Read Data by Identifier (DID read for battery voltage, software version).
- **Service 0x27:** Security Access (Seed-Key authentication).
- **Service 0x2E:** Write Data by Identifier (VIN write, end-of-line calibration).

10.2 Diagnostic Trouble Code Matrix

DTC ID	Fault Code Title	Fault Trigger Condition	Recovery Criteria	Severity
B10A2	Crash Event Recorded	Continuous LOW on PIN_03 or crash CAN pulse	Non-erasable without Security Level 27	Critical
B1001	Front Left Door Latch Jammed	Solenoid energizes for > 500ms without state change	Cycle manual door handle 3 times	Medium
B1020	Anti-Pinch Sensor Open Circuit	Analog sensor resistance > 100 kOhm	Resistance returns to 5 - 20 kOhm for 2 sec	High
B1035	Headlamp Relay High-Side Short	PROFET detects overcurrent (> 35A) during activation	Auto-retry after 30 sec cooling period	High
B1044	Rain Sensor Bus Timeout	No LIN response frame from sensor for > 500ms	Successful LIN header reply received	Low
U0100	Lost Comms with Engine ECM	CAN message 0x100 missing for > 200ms	10 consecutive valid CAN frames received	High
P0562	System Voltage Low (LVS)	Battery voltage drops below 9.0V for > 5 seconds	Battery voltage rises above 10.5V for 3 seconds	Medium

11. Verification Matrix & Traceability

11.1 Requirement to Test Case Mapping

Req ID	Requirement Description	Allocated SWC	Target Test Method
REQ-BCM-001	Doors must lock automatically when vehicle speed exceeds 15 km/h	DoorLockSWC	Hardware-in-the-Loop (HIL) CAN injection
REQ-BCM-002	All latches must unlock within 30ms of collision pulse reception	DoorLockSWC	Digital oscilloscope hardware timing verify
REQ-BCM-003	Hazard lights must oscillate at 1.5 Hz during crash condition	LightingMgrSWC	CAN and relay pulse frequency counter
REQ-BCM-004	Window express upward movement must reverse upon >100N pinch resistance	WindowCtrlSWC	Mechanical pinch rig calibrated load cell
REQ-BCM-005	System must enter Low Voltage Sleep if VBat remains < 9V for 5s	PowerMgrSWC	Programmable DC bench power supply drop test

12. Glossary and Terminology Definitions

Section 12.1: Standard Architectural Glossary

Term	Definition
SWC	Software Component: Self-contained functional unit encapsulated according to the AUTOSAR component model.
RTE	Runtime Environment: Virtual Function Bus (VFB) middleware linking SWC ports to BSW or each other.
BSW	Basic Software: Standardized software layers including Services, ECUM, Com, and MCAL drivers.
S/R Interface	Sender-Receiver Interface: Asynchronous data broadcast communication pattern between ports.
C/S Interface	Client-Server Interface: Synchronous or asynchronous operation invocation between ports.
DTC	Diagnostic Trouble Code: 3-byte identifier specifying fault classification and symptom.
ASIL	Automotive Safety Integrity Level: ISO 26262 risk classification (QM, ASIL-A through ASIL-D).
UDS	Unified Diagnostic Services: ISO 14229 protocol standard for automotive diagnostic exchange.

Section 12.2: Governance Sign-off Note

This concludes the formal High-Level Design specification for the Body Control Module. Revisions must follow formal ECN review procedures.